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Impact of breaking the language barrier on school education — Evidence from West Bengal in India

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ABSTRACT

This study examines the impact of reducing language barriers on educational outcomes in the state of West Bengal, India. Specifically, we analyze the effects of a policy reform that introduced question papers in Hindi for all subjects in grades 11 and 12 higher secondary examinations, aimed at supporting students in schools with Hindi as the language of instruction. Using school-level administrative data, we employ a difference-in-differences estimation strategy to identify the causal effects of the intervention on grade repetition and grade enrollment. Our findings reveal that the policy led to a decline in the share of repeaters and an increase in enrollment. This finding bears significant policy implications, particularly in educational settings where there is a mismatch between the language of instruction and the language used for assessment.

1. Introduction

Language is an intrinsic part of human capital and serves as a primary means of acquiring it. Language proficiency, the ability to understand and use a particular language effectively, plays a significant role in social dynamics and socioeconomic success. In societies with a majoritarian linguistic group, individuals lacking proficiency in the dominant language face significant barriers to communication and access to information, which adversely impact their social and economic outcomes. This disparity in language proficiency is referred to as a language barrier, denoting the linguistic obstacles that curtail effective communication. A large literature in economics has explored the link between language misalignment and its impact on economic outcomes. The level of proficiency in a dominant language influences returns in the labor market (Chakraborty & Bakshi, 2016; Munshi & Rosenzweig, 2006), influences one's earnings (Azam et al., 2013; Miranda & Zhu, 2013), access to healthcare (Schachter et al., 2012) and also determines outcomes in the marriage market (Miyamoto, 2022). These factors may consequently lead to social exclusion and hinder economic mobility for linguistic minority groups.

Linguistic minorities throughout the world often have to take assessment tests in a language in which they do not have full proficiency. While linguistic proficiency should be irrelevant to the test, due to limited proficiency in the language of the test, the performance of examinees is likely to be affected. In such cases, the test scores inadvertently capture the language proficiency of the candidate (Faulkner-Bond & Sireci, 2015). According to Hauser and Heubert (1999), "*if a student is not proficient in the language of the test, her performance is likely to be affected by construct-irrelevant variance — that is, her test score is likely to underestimate her knowledge of the subject being tested*" (pp. 225). Language proficiency plays a key role in the acquisition of further skills, and lack of proficiency in the dominant language can exacerbate inequality among linguistic minorities (Akresh & Akresh, 2011; Nandwani & Sen, 2025).

In our study, we evaluate the impact of language proficiency on educational outcomes by leveraging an educational policy change leading to the removal of a language barrier for a minority linguistic community in the state of West Bengal (WB) in India. The West Bengal Council of Higher Secondary Education (WBCHSE) centrally administers grade 11 and 12 examinations in schools affiliated with it. The majority of schools in West Bengal affiliated with the WBCHSE use Bengali or English as the language/medium of instruction (hereafter referred to as 'medium'). Traditionally, WBCHSE provided bilingual question papers

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[1] HS question papers in Hindi from 2018 https://www.telegraphindia.com/west-bengal/hs-question-papers-in-hindi-from-2018/cid/1395322

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0272-7757/© 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

for grade 11 and 12 exams in Bengali and English, which are also the official languages of the state. While the students of grade 11 and 12 other than Bengali and English medium schools could write their answers in their medium of instruction, they were required to comprehend the question paper in Bengali or English due to the bilingual nature of the question paper. Responding to a long-standing demand from the Hindi-speaking linguistic minority community, the WBCHSE introduced Hindi version of question papers in all subjects from the academic year 2018–19, supplementing the existing Bengali and English versions.[1] By resolving the mismatch between the medium of instruction and the language of assessment, this policy change effectively reduced language barriers faced by students in Hindi-medium schools. Using administrative school-level panel data and employing a difference-in-differences estimation strategy, we provide causal evidence on the effects of this policy on grade repetition and enrollment in Hindi-medium schools.

We find that the introduction of Hindi-language question papers led to a 2 percent reduction in the overall share of repeaters in grades 11 and 12 in Hindi-medium schools. The effect on repeaters is negative but is noisily estimated while there is a statistically significant increase in enrollment by 5.2 percent. The share of repeaters for grade 11 decreases by 3.1 percent while that for grade 12 decreases by 2.5 percent. The policy's impact is notably more pronounced for grade 11, where the number of repeaters declined by 16 percent, and total enrollment increased by 8.6 percent. The results suggest that language barriers at the entry point to higher secondary education may be contributing to the observed change.

Studies have shown that a mismatch between the language of instruction and students' mother tongue affects literacy and college graduation rates (Jain, 2017), learning outcomes (Datta & Khatri, 2023), grade repetition (Tagat & Samaddar, 2025) and grade points (Bernhofer & Tonin, 2022). While the impact of mismatch between the language of instruction and mother tongue is common, other forms of language mismatch have been less explored. For linguistic minorities worldwide, a mismatch between the language of test assessment and their limited proficiency in that language is a common challenge. There is a small but growing literature in Economics on the impact of language proficiency on educational outcomes (Akresh & Akresh, 2011; Cavallo & Russo, 2025; Fenoll, 2018; Fenoll & Kuehn, 2025; Isphording et al., 2016; Mukhopadhyay, 2020). Akresh and Akresh (2011) and Mukhopadhyay (2020) find that limited proficiency in the language of test assessment adversely affected the performance of students in standardized language tests. There is also some evidence regarding the impact of language proficiency on math test scores, although the findings are mixed. Fenoll (2018) and Mukhopadhyay (2020) did not find evidence in support of the hypothesis that language proficiency impacts performance in math tests. However, contrary to this Akresh and Akresh (2011), Isphording et al. (2016) and Cavallo and Russo (2025) find that lack of linguistic proficiency in the language of test assessment adversely affects performance in math tests. Recently, Fenoll and Kuehn (2025) finds that linguistic proficiency may or may not affect math test scores, depending on whether proficiency is measured as oral language ability (i.e., the ability to speak or understand the language) or includes proficiency in reading and writing as well.

Estimating the causal impact of language proficiency on test performance is challenging, as it is difficult to disentangle the effect of language proficiency from confounding factors, such as ability and motivation, which influence both language proficiency and test performance. Hence, much of the existing literature in economics focuses primarily on immigrants, using age at arrival in the host country as a source of exogenous variation to causally identify the impact of language proficiency on test scores. While the question of how language proficiency impacts educational performance is highly relevant for the immigrant population as they acquire human capital in the host country, it is equally pertinent to other non-immigrant linguistic minorities who lack proficiency in the dominant language. Our paper contributes to the literature by measuring the causal impact of language proficiency on educational outcomes among a non-immigrant linguistic minority community, using a difference-in-differences strategy.

In our setting, Hindi medium students have limited proficiency in Bengali or English. Therefore, we expect that the policy change from bilingual to trilingual question papers through the inclusion of Hindi language versions would have positively impacted their test performance. It may appear that the policy change is too modest to significantly affect test performance, considering that students had always been permitted to write their answers in their medium of instruction. However, findings from cognitive neuroscience and psycholinguistic literature suggest that language switching imposes cognitive costs, particularly when the language used for information retrieval differs from the language of learning — even for students who are perfectly bilingual. This mismatch increases cognitive load and can hinder performance (Bernhofer & Tonin, 2022).

The impact of language proficiency on human capital acquisition can have profound implications on linguistic minorities in very diverse multilingual countries with one or few dominant languages. In such settings, it is not uncommon to see that students from linguistic minorities often have to take an assessment test in a language in which they do not have full proficiency. When scores of these standardized tests are used to evaluate the ability of the students for important economic outcomes like higher education and jobs, it may further compound the disadvantages of the linguistic minority community. In a recent policy paper on effective language of instruction policies for learning, the World Bank strongly recommended aligning the language of assessment with the language of instruction (Crawford & Marin, 2021). Therefore, our results have policy implications for addressing the language barrier that exists in the form of students taking test assessments in languages in which they have limited proficiency. While it may be practically challenging for policymakers to conduct tests and assessments in languages in which the students are most proficient, there may be significant gains from this exercise.

The remainder of this paper is organized as follows. In Section 2, we discuss the background setting of the education policy along with the data used in our study. The empirical strategy is outlined in Section 3. The results are presented in Section 4, with robustness checks provided in Section 5. Section 6 provides a discussion of the findings and its policy implications. Finally, Section 7 offers concluding remarks.

2. Background and data

The school education system in India follows a 10+2 structure. School education is governed by various central and state educational agencies called education boards, which are administrative bodies responsible for administering the curriculum and exams in schools affiliated with them. All schools in the country are affiliated to one of the central/state boards. The education board also determines the language or the medium of instruction in schools. For example, among the two major all India boards — the Central Board of Secondary Education (CBSE) and the Council for the Indian School Certificate Examinations (CISCE), the former mandated English or Hindi as the only medium of instruction,[2] whereas the latter had only English medium schools affiliated to it. On the other hand, schools affiliated with most state education boards have different languages as the medium of instruction depending upon the linguistic diversity of the state, of which the majority of schools use the dominant language in the state as the medium of instruction. While schools affiliated with CBSE and CISCE are spread across the country, state board schools are mostly restricted to their respective states.

[2] CBSE allowed schools to teach in Indian languages other than English and Hindi as the only medium of instruction since 2023 https://timesofindia.indiatimes.com/india/cbse-schools-can-now-teach-in-local-languages/articleshow/102026894.cms.

Table 1
Medium of instruction and Board of Education.

Medium of instruction	WBCHSE	Other Boards	Total
Bengali	75.6%	0.2%	75.8%
English	1.2%	2.4%	3.6%
Bengali and English	2.4%	0.1%	2.5%
Hindi	1.9%	0.0%	1.9%
Others	0.8%	0.0%	0.9%
Multiple mediums	14.5%	0.9%	15.2%
Total (N = 7213)	**96.5%**	**3.5%**	**100.0%**

Notes: The table reports the percentage of schools out of a total of 7213 schools in 2017 with grades 11 and 12 that were affiliated to WBCHSE and other boards along with their medium of instruction.

The West Bengal Council of Higher Secondary Education (WBCHSE) is a state education board in the state of West Bengal (WB) that acts as a nodal body for conducting the final examinations for grades 11 and 12 for all schools affiliated with the WBCHSE. As of 2017, there were 7213 schools in WB with grades 11 and 12 affiliated to different boards. Among these, 75.6 percent of schools had Bengali as the sole medium of instruction and were affiliated with WBCHSE (as reported in Table 1). The remaining schools had some other language as the medium of instruction or had multiple mediums of instruction.[3] This predominance of Bengali-medium schools mainly reflects the linguistic landscape of the state, where Bengali serves as the native language for 86.43% of the state's population. Other significant linguistic communities in WB include Hindi (6.98%), Santali (2.67%), Urdu (1.83%), and Nepali (1.27%) (Census, 2011).

The academic session for grades 11 and 12 in schools affiliated with the WBCHSE board typically starts in May and ends with final exams in March or April of the subsequent year. For the rest of our paper, we refer to the year as the year of the start of an academic session such that 2015 would mean the 2015–16 academic year, with final exams being conducted in March or April of 2016. The final examinations for grades 11 and 12 are standardized exams since WBCHSE centrally administers them, and all students in grades 11 and 12 in schools affiliated to WBCHSE across the state receive the same question paper in the final exam.[4] However, the answer sheets of grade 12 are checked anonymously by external examiners unlike grade 11, which are checked by school teachers internally even when the question papers for grade 11 were provided centrally by the WBCHSE. Prior to the academic session 2017–2018, WBCHSE provided bilingual question papers for grade 11 and 12 exams, which were available only in Bengali and English. Although all students were allowed to write answers in the language of their medium of instruction, this posed a challenge for students studying in mediums other than Bengali and English as they had to comprehend the questions in a language with limited proficiency. The demand for questions in Hindi for Higher Secondary education first surfaced in the late 1990s during the erstwhile Left regime. Similar appeals by students and teachers of Hindi-medium schools were made to the state government during the current regime since 2011. In 2018, the WBCHSE[5] finally introduced question papers in Hindi for all subjects in grades 11 and 12.

[3] A school with multiple mediums of instruction means that the same school had separate sections of the same grade using different language as the medium of instruction. Therefore, a school with multiple mediums does not imply a multi-lingual medium of instruction.

[4] WBCHSE decided to decentralize exams for grade 11 from the academic year 2024–25 onward so that each school will conducting their own exams in class 11 https://www.thehindu.com/news/cities/kolkata/west-bengal-schools-to-conduct-their-own-class-xi-exams/article66892640.ece

[5] During the middle of the 2017 academic year, the WBCHSE, for the first time, announced the provision of Hindi versions of question papers for grades 11 and 12. However, due to limited time, implementation was restricted to only a few subjects.

Our study evaluates the impact of the above policy change on the educational outcomes for Hindi medium students. We used administrative data from the Unified District Information System for Education (UDISE) for years 2015, 2016, and 2017, as well as its updated version, UDISE+, for years 2020 and 2021. This database covers school-level information on grade-wise enrollment and repetition, medium of instruction, the school's demographic details, and the school's infrastructure, as well as information on teachers in the school. The medium of instruction variable is the primary variable that we used to identify our treatment and control groups. Our treatment group is the Hindi medium schools, and the control group consists of schools with Bengali or English or both Bengali and English as the only two mediums of instruction. Henceforth, we refer to our control group schools as Bengali and English medium. The data set provides information on all variables for each school aggregated across all mediums of instruction. Therefore, if a school has more than one medium of instruction, we do not observe the variables separately for each medium of instruction in that school. Therefore, we excluded schools that had multiple mediums of instruction (except those that had only Bengali and English medium). We also excluded schools whose sole medium of instruction was a language other than Bengali, English, or Hindi. The policy change ensured that students in the control group schools (Bengali and English medium schools) would not be affected, as they would continue to comprehend questions from the English and Bengali versions of the question paper. However, after the policy change, students in the treatment group (Hindi medium schools) now have access to the Hindi version of the question paper in which they are proficient, as Hindi has been their medium of instruction. Therefore, the policy is expected to improve the performance of the students in the treatment group.

We have balanced panel data for 1406 schools in the control group and 63 schools in the treatment group spread across 11 districts of West Bengal (as in A.1). Districts without any Hindi-medium schools were excluded from our analysis, which means our control and treatment units are from the same districts. Table A.1 in the appendix provides district-wise distribution of treatment and control schools. Moreover, due to the unavailability of data on the number of repeaters for the 2018 and 2019 cohorts, we could not include them in our analysis. Given the policy was implemented in 2018, the data from 2015–2017 serves as the pre-intervention period, while the data for 2020 and 2021 accounts for the post-intervention period. Table 2 provides the summary statistics of all variables used in the study. Column (1) reports the mean and standard deviation of all the variables used in the study for the overall sample. Columns (2) and (4) report the mean and standard deviation for the treated schools (Hindi medium) in the pre-intervention and post-intervention periods, respectively, while columns (3) and (5) report for the control schools (Bengali/English) in the pre-intervention and post-intervention periods respectively. The percentage of repeaters in the treated schools falls from 2.7 percent in the pre-intervention period to 0.3 percent in the post-treatment period, while that in the control schools remains relatively flat at 1 percent throughout.

3. Empirical strategy

To evaluate the impact of the policy on educational outcomes, we employ a difference-in-difference (DID) estimation strategy given below in (1).

$$Y_{idt} = \alpha + \beta(\text{Treat}_i \times \text{Post}_t) + \theta X_{idt} + \gamma_i + \tau_t + \epsilon_{idt} \quad (1)$$

Y_{idt} is the outcome variable of interest for school i, in district d, and year t. Treat_i is the treatment indicator, which takes the value of 1 for schools with Hindi as the medium of instruction and 0 for Bengali and English medium schools under WBCHSE. Post_t is the intervention dummy that takes the value 1 for the post-intervention years of 2020 and 2021 and 0 for the pre-intervention years of 2015, 2016 and 2017. X_{idt} includes the school-level control variables that vary with

Table 2
Summary statistics.

Variables	Full sample	Pre intervention		Post intervention	
		Treated	Control	Treated	Control
	(1)	(2)	(3)	(4)	(5)
Share of repeaters	0.014	0.027	0.011	0.003	0.017
	(0.084)	(0.145)	(0.062)	(0.008)	(0.107)
Number of repeaters	3.96	12.45	2.96	0.99	5.02
	(30.84)	(77.61)	(19.55)	(2.75)	(38.74)
Number of enrollment	254.76	346.58	245.42	382.53	256.86
	(152.7)	(192.67)	(145.93)	(209.15)	(152.01)
School characteristics					
Prop. of schools with electricity	0.991	0.968	0.988	1.000	0.998
	(0.094)	(0.176)	(0.110)	(0.000)	(0.050)
Prop. of schools approachable by all weather road	0.946	0.974	0.938	0.984	0.953
	(0.227)	(0.161)	(0.241)	(0.125)	(0.211)
Number of classrooms in good condition	12.778	9.868	10.980	15.294	15.557
	(7.526)	(6.974)	(6.805)	(10.462)	(7.546)
Number of functional toilets	6.358	7.952	5.909	9.571	6.782
	(4.829)	(7.850)	(4.173)	(8.656)	(5.120)
Number of teachers	25.325	20.651	25.622	21.111	25.383
	(9.190)	(9.496)	(9.153)	(10.328)	(9.045)
Percentage of teachers who are graduate	90.639	99.189	99.086	73.755	78.151
	(22.455)	(4.957)	(4.471)	(34.893)	(30.836)

Notes: Summary statistics is the mean of the variable unless mentioned with standard deviations reported in parentheses. Column(1) reports the summary statistics for the overall sample. The sample consists of a balanced panel of 63 Hindi medium schools in the treatment group and 1406 Bengali/English medium schools in the control group. Columns (2) and (4) reports the summary statistics for the treatment group in the pre-intervention period and the post-intervention period respectively. Columns (3) and (5) reports the summary statistics for the control group in the pre-intervention period and the post-intervention period respectively. The pre-intervention years are 2015, 2016 and 2017 while the post-intervention years are 2020 and 2021. The variable "Share of repeaters" is defined as the ratio of the number of repeaters to the total enrollment in grades 11 and 12.

time - electricity connection at the school, school accessibility by road, number of good classrooms in the school, number of functional toilets in the school, total number of teachers, and percentage of school teachers who are graduate and above. We account for academic cohort-specific variations by including year fixed effects τ_t, ruling out the need to include a post-intervention fixed effect dummy separately. γ_i denotes the school fixed effects, which take into account any unobserved, time-invariant characteristics of schools that can affect the outcome variable. Hence, we do not include a treatment dummy separately in Eq. (1) as it is absorbed in the school fixed effects. β is our coefficient of interest, which estimates the causal impact of the policy intervention. We cluster standard errors at the school level.

Our primary outcome of interest is the share of repeaters given by:

$$Share\ of\ Repeaters = \frac{Total\ Number\ of\ Repeaters}{Total\ Number\ of\ Enrollment}$$

We use the natural logarithm of this share as the primary dependent variable. However, to account for schools that have zero share of repeaters, we add one to the share. If the intervention had an effect, we should expect a decline in the share of repeaters in Hindi medium schools. This would be due to a decrease in the total number of repeaters in Hindi medium schools, as the provision of the Hindi version of question papers would have helped them comprehend and understand questions better, potentially reducing the number of repeaters. Additionally, the policy might have encouraged higher student enrollment in grades 11 and 12 in Hindi medium schools if it boosted the confidence of students in taking exams in Hindi medium schools by removing the language barrier. Therefore, we also evaluate the impact of the policy on the number of repeaters and student enrollment.

Our post-intervention years coincide with the outbreak of COVID-19 pandemic, which may affect our outcome variable. The time-fixed effects included in Eq. (1) should account for this impact. However, the intensity of the outbreak of COVID-19 may not have been uniform across the districts in India (Agarwal & Naha, 2023; Gangwar & Ray, 2021) and hence can affect the outcome variable differently depending upon the district in which the school is located. Hence, we additionally control for annual district COVID-19 mortality rate in Eq. (1). The COVID-19 mortality rate is defined as the ratio of the number of deaths due to COVID-19 to the total number of confirmed cases in a particular year.[6] The data for the district-wise COVID-19 annual mortality rate is sourced from the district level Covid-19 India data set.[7] In an alternate estimation strategy, we control for district-by-year fixed effects to account for time-specific unobserved variables that may influence the outcome.

4. Results

Main results. Table 3 reports the impact of the policy on the share of repeaters in Hindi medium schools under WBCHSE. We observe that the estimate of β is negative and statistically significant at 1% level. Column (1) shows that the share of repeaters fell by 2 percent after controlling for time-varying school characteristics along with school and year fixed effects. In column (2), we additionally control for the annual district COVID-19 mortality rate, and the estimate of β remains unchanged. In Column (3), we estimate Eq. (1) with district-by-year fixed effects, and the estimate of β is still negative and statistically significant with a treatment effect of 2.1 percent.

However, this decrease in the share of repeaters could be driven by a decrease in the number of repeaters, an increase in the number of student enrollments, or both. Table 4 reports the impact of the policy on the natural log of repeaters as the outcome variable,[8] after controlling for the total enrollment. Columns (1)–(3) in Table 4 report the same specifications as in Table 3. The estimates in columns (1)–(3) in Table 4 are negative but statistically insignificant. The coefficient in all the specifications is around −0.1, indicating that, on average, the policy led to a decline of repeaters by 10%, but this result is not statistically significant. Hence, we cannot confidently conclude that the policy had

[6] The value for COVID-19 mortality rate has been set to zero for all the districts in the pre-covid years i.e. 2015, 2016 and 2017, which are also our policy pre-intervention years.

[7] Accessed from https://www.kaggle.com/datasets/datathon2021/covid19-india-dataset

[8] To account for schools that have zero repeaters, we add one to the number of repeaters.

a significant impact on repeaters after controlling for enrollment. Table 5 reports the impact of the policy on the natural log of enrollment as the outcome variable. Columns (1)–(3) in Table 5 report the same specifications as in Table 3. In columns (1) and (2), the estimate of β is positive and statistically significant at 10% level. The estimates show that the introduction of the policy led to an increase in enrollment by around 5 percent. In column (3), with district-by-year fixed effects, the estimate, even though it is positive, is statistically insignificant. Our overall results indicate that the policy led to a decline in the share of repeaters, and this impact is primarily driven by greater enrollment. We also report the impact of the policy on the share of repeaters, the number of repeaters (controlling for enrollment), and the number of enrollment in levels in Table A.2 in the appendix.

Table 6 reports the impact of the policy on the share of repeaters, the number of repeaters, and the number of enrollment for grades 11 and 12 separately, where we use the natural log of the three variables as our dependent variable. Column (1) and (4) shows that the introduction of question papers in Hindi led to a decline in the share of repeaters by 3.1 percent for students in grade 11 and 2.5 percent in grade 12, respectively. Column (2) and (5) shows that the policy led to a decline in repeaters by 16 percent for grade 11 and 6.5 percent for grade 12, respectively. However, the co-efficient for grade 12 in column (5) is statistically insignificant, implying no significant impact for grade 12. Column (3) reports that the policy had a positive and statistically significant impact on enrollment for grade 11, with an effect size of 8.6 percent. The corresponding coefficient for enrollment in grade 12 in column (6) is small and statistically insignificant, implying no significant impact. Hence, we find that the impacts are more pronounced for grade 11 as compared to grade 12, suggesting that the language barrier at the entry point to higher secondary school education might be contributing to the observed effect.

The more pronounced effect of the policy on grade 11 can be attributed to one of the following factors. Firstly, higher secondary education in grades 11 and 12 has a steep learning curve compared to grade 10 secondary school education. The language barrier can further compound the level of difficulty of education for students in Hindi medium schools as they encounter it for the first time in grade 11. In our period of study, the WBCHSE was responsible for preparing question papers for the annual examinations of grade 11 in schools and the higher secondary (HS) exam at the end of grade 12 as well. Unlike in grades 11 and 12, there was no mismatch in the medium of instruction and question papers in grade 10, since the Hindi version of the question papers are provided in exams of grade 10. Therefore, we possibly observe a stronger impact of the policy in grade 11 compared to grade 12 as students adapt between grades 11 and 12. Secondly, schools often use strict screening mechanisms in advancing students from grade 11 to 12. Students failing in grade 12 exams bring a bad reputation to a school since the performance of students in grade 12 exams is considered a metric of the school's performance. Therefore, schools follow strict screening of students in grade 11 while allowing students to progress from grade 11 to 12 only if they do not doubt the capability of the student to be fit for class 12.

Validity of parallel trend assumption. The identifying assumption underlying our analysis and making a causal claim about our results is that in the absence of the policy, the share of repeaters in the Hindi medium schools (treatment group) would have continued in parallel to trends in the share of repeaters in Bengali/English medium schools (control group) after 2018. However, this is an assumption about the counterfactual and, therefore, is not fundamentally testable. Therefore, we first attempt to see if the trends in the outcome variables for the pre-policy years are similar for the treated and control schools. Fig. 1 plots the year-wise averages of the share of repeaters (in natural log)[9] by treatment status. Figs. A.1, A.2, and A.3 in the appendix plots the year-wise average of the share of repeaters, number of repeaters, and total enrollment (in levels) for the control and treated schools. As shown in Fig. 1, the parallel trends assumption may be violated. The observed pattern in the share of repeaters in Fig. A.1 in the appendix is primarily driven by changes in the absolute number of repeaters, as shown in Fig. A.2. As seen in Fig. A.3 in the appendix, the trend in enrollment across treated and control schools reveals that enrollment in Bengali/English medium schools under WBCHSE remained relatively constant throughout the study period. However, enrollment in Hindi medium schools under WBCHSE shows an increasing trend following the policy change.

Next, we follow the usual practice to test for the parallel trend assumption in quasi-experimental studies and test the hypothesis of parallel pre-trends for the period 2015, 2016, and 2017 by restricting our sample to the pre-intervention period only. We assign a false treatment status to a pre-intervention year and then estimate the treatment effect of this false treatment using Eq. (1) for our primary outcome variable. For the parallel trend assumption to be satisfied, the co-efficient β should be insignificant, suggesting that the treated and control schools experienced similar trends in the pre-treatment period as compared to the reference false treatment year. Columns (1) and (2) in Table A.3 in the appendix present the results on the share of repeaters (in natural log) with 2016 and 2017 as the false treatment year, respectively. We find no significant effect when the false treatment year is 2017 (column (2)), but the coefficient in column (1) is statistically significant but positive when both 2016 and 2017 are assigned as the false treatment years, suggesting the presence of pre-trends.

Table 3
Impact of question papers in Hindi on share of repeaters.

	ln(Share of repeaters)		
	(1)	(2)	(3)
Treat × Post	−0.020***	−0.020***	−0.021***
	(0.007)	(0.007)	(0.007)
Treat			0.011
			(0.007)
Effect size (%)	2.0	2.0	2.1
Observations	7345	7345	7345
Adjusted R^2	0.01	0.01	0.49
School controls	Yes	Yes	Yes
District covid mortality rate	No	Yes	No
School FE	Yes	Yes	No
Year FE	Yes	Yes	No
District × Year FE	No	No	Yes

Notes: This table reports the difference-in-difference estimates of the impact of the introduction of question papers in Hindi on the share of repeaters. The dependent variable is the natural log of the number of repeaters as a ratio of the total enrollment for grades 11 and 12 combined. The sample consists of a balanced panel of 1406 Bengali/English medium schools under WBCHSE in the control group and 63 Hindi medium schools under WBCHSE in the treatment group. Treat is an indicator variable that takes a value of 1 for Hindi medium schools and 0 for Bengali/English medium schools. Post is an indicator variable that takes a value of 1 for the post-intervention years of 2020 and 2021 and 0 for the pre-intervention years of 2015, 2016, and 2017. All regressions include school-level controls that vary with time - electricity connection at the school, school accessibility by road, number of good classrooms in the school, number of functional toilets in the school, total number of teachers, and percentage of school teachers who are graduates. Column (2) additionally controls the annual COVID mortality rate at the district level (collected from the website https://www.kaggle.com/datasets/datathon2021/covid19-india-dataset.) The annual COVID mortality rate is calculated as the share of deaths to the total number of confirmed cases due to COVID-19 in a particular year and is equal to zero for the years 2015, 2016, and 2017. Columns (1) and (2) include school and year fixed effects, while column (3) includes only district-by-year fixed effects. Standard errors are clustered at the school level and reported in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

[9] To account for schools that have zero share of repeaters, we add one to the share.

Table 4
Impact of question papers in Hindi on repeaters.

	ln(Repeaters)		
	(1)	(2)	(3)
Treat × Post	−0.107	−0.106	−0.102
	(0.0114)	(0.113)	(0.109)
Treat			−0.078
			(0.090)
Effect size (%)	10.7	10.6	10.2
Observations	7345	7345	7345
Adjusted R^2	0.18	0.18	0.09
School controls	Yes	Yes	Yes
District covid mortality rate	No	Yes	No
School FE	Yes	Yes	No
Year FE	Yes	Yes	No
District × Year FE	No	No	Yes

Notes: This table reports the difference-in-difference estimates of the impact of the introduction of question papers in Hindi on repeaters. The dependent variable is the natural log of the number of repeaters in grades 11 and 12 combined. The sample consists of a balanced panel of 1406 Bengali/English medium schools under WBCHSE in the control group and 63 Hindi medium schools under WBCHSE in the treatment group. Treat is an indicator variable that takes a value of 1 for Hindi medium schools and 0 for Bengali/English medium schools. Post is an indicator variable that takes a value of 1 for the post-intervention years of 2020 and 2021 and 0 for the pre-intervention years of 2015, 2016, and 2017. All regressions include school-level controls that vary with time - number of students enrolled, electricity connection at the school, school accessibility by road, number of good classrooms in the school, number of functional toilets in the school, total number of teachers, and percentage of school teachers who are graduates. Column (2) additionally controls the annual COVID mortality rate at the district level (collected from the website https://www.kaggle.com/datasets/datathon2021/covid19-india-dataset.) The annual COVID mortality rate is calculated as the share of deaths to the total number of confirmed cases due to COVID-19 in a particular year and is equal to zero for the years 2015, 2016, and 2017. Columns (1) and (2) include school and year fixed effects, while column (3) includes only district-by-year fixed effects. Standard errors are clustered at the school level and reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

Table 5
Impact of question papers in Hindi on enrollment.

	ln(Enrollment)		
	(1)	(2)	(3)
Treat × Post	0.052*	0.053*	0.029
	(0.031)	(0.031)	(0.041)
Treat			0.6532***
			(0.067)
Effect size (%)	5.2	5.3	2.9
Observations	7345	7345	7345
Adjusted R^2	0.88	0.88	0.49
School controls	Yes	Yes	Yes
District covid mortality rate	No	Yes	No
School FE	Yes	Yes	No
Year FE	Yes	Yes	No
District × Year FE	No	No	Yes

Notes: This table reports the difference-in-difference estimates of the impact of the introduction of question papers in Hindi on enrollment. The dependent variable is the natural log of the number of enrollment in grades 11 and 12 combined. The sample consists of a balanced panel of 1406 Bengali/English medium schools under WBCHSE in the control group and 63 Hindi medium schools under WBCHSE in the treatment group. Treat is an indicator variable that takes a value of 1 for Hindi medium schools and 0 for Bengali/English medium schools. Post is an indicator variable that takes a value of 1 for the post-intervention years of 2020 and 2021 and 0 for the pre-intervention years of 2015, 2016, and 2017. All regressions include school-level controls that vary with time - electricity connection at the school, school accessibility by road, number of good classrooms in the school, number of functional toilets in the school, total number of teachers, and percentage of school teachers who are graduates. Column (2) additionally controls the annual COVID mortality rate at the district level (collected from the website https://www.kaggle.com/datasets/datathon2021/covid19-india-dataset.) The annual COVID mortality rate is calculated as the share of deaths to the total number of confirmed cases due to COVID-19 in a particular year and is equal to zero for the years 2015, 2016, and 2017. Columns (1) and (2) include school and year fixed effects, while column (3) includes only district-by-year fixed effects. Standard errors are clustered at the school level and reported in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.1$.

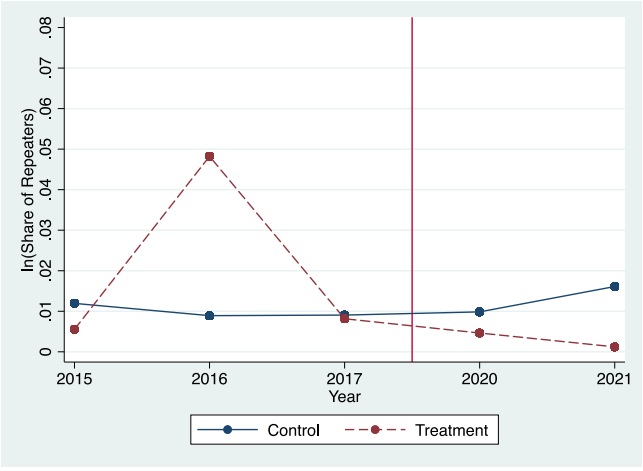


Fig. 1. Year-wise average of ln(Share of Repeaters) by treatment status
Notes: The above figure presents the average of the ln(share of repeaters) over time for the control schools (Bengali/English medium) and the treated schools (Hindi medium). The vertical line denotes the implementation of the policy.

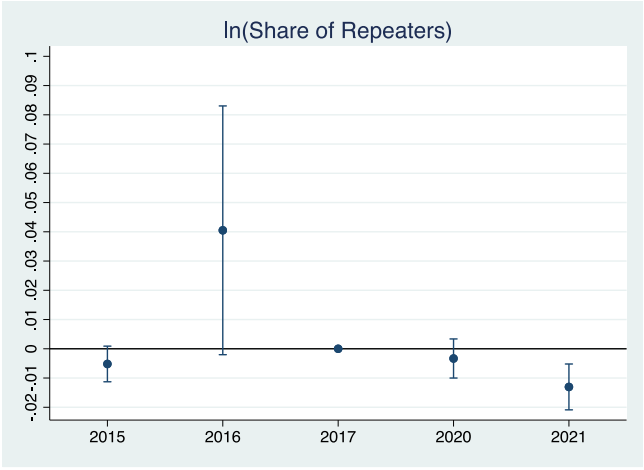


Fig. 2. Event study estimates for the effect of the introduction of question papers in Hindi on share of repeaters.
Notes: The dependent variable is the natural log of the number of repeaters as a ratio of the total enrollment for grades 11 and 12 combined. The sample consists of a balanced panel of 1406 Bengali/English medium schools under WBCHSE in the control group and 63 Hindi medium schools under WBCHSE in the treatment group. The regression includes school-level controls that vary with time - electricity connection at the school, school accessibility by road, number of good classrooms in the school, number of functional toilets in the school, total number of teachers, and percentage of school teachers who are graduates. It also includes school and year fixed effects, and standard errors are clustered at the school level.

To further examine the validity of our difference in difference framework, we estimate the following event study specification

$$Y_{idt} = \alpha + \sum_{s \neq \{2017\}} \beta_s(\text{Treat}_i \times \text{Year}_s) + \theta X_{idt} + \gamma_i + \tau_t + \epsilon_{idt} \quad (2)$$

where $s \in \{2015, 2016, 2020, 2021\}$ and the reference year is 2017, the year prior to the implementation of the policy. Hence, the magnitude of the coefficients can be interpreted as the change in the difference

Table 6
Impact of question papers in Hindi on repeaters and enrollment: grade-wise.

	Grade 11			Grade 12		
	ln(Share of repeaters) (1)	ln(Repeaters) (2)	ln(Enrollment) (3)	ln(Share of repeaters) (4)	ln(Repeaters) (5)	ln(Enrollment) (6)
Treat × Post	−0.031***	−0.160*	0.086***	−0.025**	−0.065	0.014
	(0.011)	(0.092)	(0.030)	(0.01)	(0.09)	(0.04)
Effect size (%)	3.1	16.0	8.6	2.5	6.5	1.4
Observations	7345	7345	7345	7345	7345	7345
Adjusted R^2	0.01	0.15	0.82	0.01	0.14	0.84
School controls	Yes	Yes	Yes	Yes	Yes	Yes
School FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Columns (1), (2) and (3) report the difference-in-difference estimates of the impact of the introduction of question papers in Hindi for grade 11 while columns (4), (5) and (6) report the results for grade 12 respectively. The dependent variable in columns (1) and (4) is the natural log of the share of repeaters, which is defined as the number of repeaters as a ratio of the total enrollment. The dependent variable in columns (2) and (5) is the natural log of the number of repeaters and the dependent variable in columns (3) and (6) is the natural log of the number of enrollment. The sample consists of a balanced panel of 1406 Bengali/English medium schools under WBCHSE in the control group and 63 Hindi medium schools under WBCHSE in the treatment group. Treat is an indicator variable that takes a value of 1 for Hindi medium schools and 0 for Bengali/English medium schools. Post is an indicator variable that takes a value of 1 for the post-intervention years of 2020 and 2021 and 0 for the pre-intervention years of 2015, 2016, and 2017. All regressions include school-level controls that vary with time - electricity connection at the school, school accessibility by road, number of good classrooms in the school, number of functional toilets in the school, total number of teachers, and percentage of school teachers who are graduates. Column (2) and Column (5) additionally controls the total enrollment. All regressions include school and year fixed effects. Standard errors are clustered at the school level and reported in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

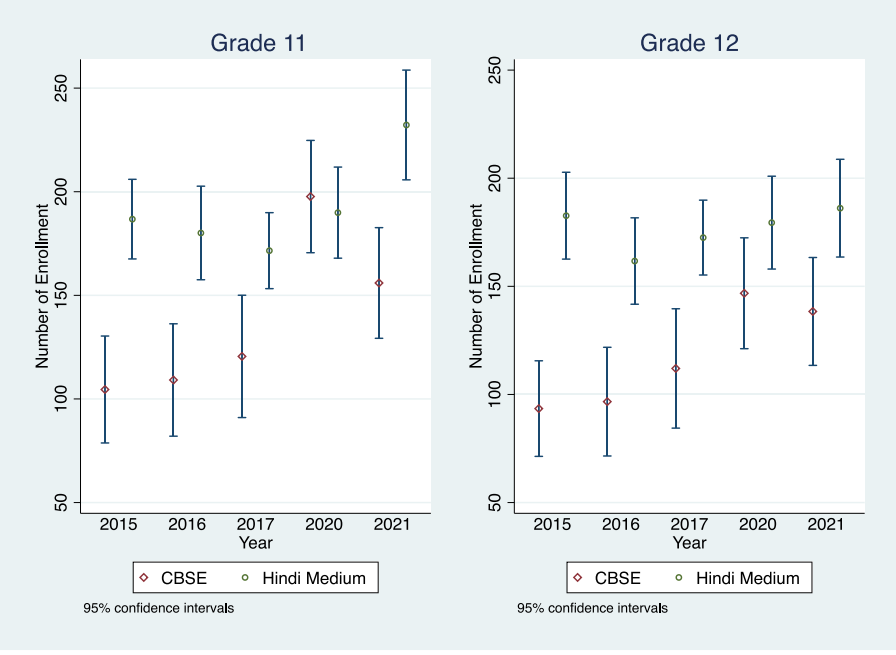


Fig. 3. Gradewise average number of enrollment across CBSE and Hindi medium schools.
Notes: The above figure presents the average number of enrollment over time for Hindi medium schools under WBCHSE and schools under CBSE for grades 11 and 12 separately. The sample includes an unbalanced panel of 233 CBSE schools and 159 Hindi medium schools under WBCHSE.

between treated and control groups relative to the 2017 difference. We use the same set of time-varying school-level controls and include school and year fixed effects. We cluster the standard errors at the school level. For the parallel pre-trends to hold, we must obtain β_{2015} and β_{2016} to be close to zero or be statistically insignificant. Fig. 2 plots the events study graph corresponding to Eq. (2) with the natural log of share of repeaters as the dependent variable.

In summary, the parallel trends assumption appears to be violated in our case, indicating that the difference-in-differences estimates may be biased. To address this concern, we next provide estimates of the impact of the policy on our outcome variables using the synthetic differences-in-differences approach following Arkhangelsky et al. (2021).

Synthetic differences-in-differences. This approach uses a weighted average of the control units to serve as the counterfactual for the treated units, offering potential improvements over standard difference-in-difference models. Importantly, this reweighting is conditioned on pre-intervention outcomes to ensure a more comparable construction of the counterfactual. The results are reported in Table 7. Column (1) reports the impact of the policy on the share of repeaters, while columns (2) and (3) report the estimates for the number of repeaters and enrollment, respectively. In all the regressions, we consider the natural log of the outcome variable. Column (1) shows that the policy led to a decline in the share of repeaters by 1 percent, which is similar to our DID estimates in Table 3. Column (2) reports no impact of the policy on repeaters after controlling for enrollment, consistent with our DID estimates in Table 4. Column (3) shows an 8 percent increase in enrollment following the policy implementation. The coefficient is statistically significant at the 1% level and is larger in magnitude compared to the DID coefficients reported in Table 5. For our synthetic difference-in-differences models, we report bootstrapped standard errors based on 500 replications. Our synthetic difference-in-differences model yields parallel pre-trends in the counterfactual by construction, and we present these plots in Fig. A.4 in the appendix.

Placebo test with grade 10. To establish the validity of our results, we replicate our analysis for the share of repeaters in grade 10. Grade 10 Hindi medium schools should remain unaffected since the policy was implemented only for grades 11 and 12 in WBCHSE schools, while Class 10 students fall under a different state educational council, the West Bengal Board of Secondary Education (WBBSE). Moreover, question papers for exams in grade 10 had been provided in Hindi, in addition to Bengali and English for a long time. So, using outcomes of grade 10 students as a placebo treatment group, we estimate (1) and present the results in column (3) of Table A.3 in the appendix. We find no impact of the policy on the share of repeaters (in natural log) for grade 10.

Simulations with random assignment to treatment. Following Bharadwaj et al. (2014), we also performed another placebo test by randomly assigning treatment status to schools while ensuring that the sample size in our new treatment and control group matched the actual composition. We estimate the interaction term, β in Eq. (1) for this placebo treatment. We repeat this exercise 1,000 times, each time randomly assigning treatment and control status to schools in our sample. The intuition behind this exercise is that if our identification strategy picks up the true effect, we would then expect that most of the coefficients in the simulated regressions would be insignificant and also small in magnitude.

Fig. A.5 in the appendix shows the density plot of the 1000 simulated coefficients for our primary outcome variable, i.e., the natural log of the share of repeaters. We find that the distribution of these coefficients is concentrated at zero and far from our actual estimate, which is depicted through a vertical line. The percentage of false estimate coefficients that are statistically significant at 5 percent level is only 14.7%. This suggests that the estimation of the actual treatment effect is unlikely to arise due to chance.

5. Robustness checks

Alternative control group. English medium schools predominantly cater to students belonging to relatively well-off backgrounds and are significantly different from Bengali and Hindi medium schools, as can be seen in differences in school infrastructure in Table A.4 in the appendix. We exclude them from our control group to ensure a more comparable analysis with only the Bengali and Hindi medium schools and estimate Eq. (1) with only Bengali medium schools in the control group. Column (1) in Table A.5 in the appendix reports the result for the natural log of the share of repeaters, and the finding is comparable to our main result in column(1) in Table 3.

Table 7
Synthetic difference-in-difference estimates of the impact of question papers in Hindi on repeaters and enrollment.

	ln(Share of Repeaters) (1)	ln(Repeaters) (2)	ln(Enrollment) (3)
Treat × Post	−0.01**	−0.14	0.08***
	(0.003)	(0.120)	(0.025)
Effect size (%)	1.0	12.0	8.0
Observations	7345	7345	7345
School FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Notes: This table presents estimates from the synthetic difference-in-differences model outlined in Arkhangelsky et al. (2021). The dependent variable in column (1) is the natural log of the share of repeaters, which is defined as the number of repeaters as a ratio of the total enrollment in grades 11 and 12. The dependent variable in column (2) is the natural log of the number of repeaters in grades 11 and 12, and the dependent variable in column (3) is the natural log of the number of enrollment in grades 11 and 12. Treat is an indicator variable that takes a value of 1 for Hindi medium schools and 0 for Bengali/English medium schools. Post is an indicator variable that takes a value of 1 for the post-intervention years of 2020 and 2021 and 0 for the pre-intervention years of 2015, 2016, and 2017. All regressions include school and year fixed effects. Bootstrapped standard errors based on 500 simulations are presented in parentheses. Additionally, the specification in column (2) is conditioned on the covariate – number of students enrolled – using the projected approach (Kranz, 2022). ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

Treatment year adjustment. In December 2017, when WBCHSE decided to implement the policy, question papers in very few subjects were introduced in Hindi for the first time. Therefore, we reestimated Eq. (1) after excluding this year from the analysis. We now have only 2015 and 2016 as the pre-intervention years, and 2020 and 2021 as the post-intervention years. Column (2) in Table A.5 in the appendix reports the results for the natural log of the share of repeaters. The result is similar in magnitude compared to our main result in column(1) in Table 3.

6. Discussion

In this section, we discuss the potential policy implications of our research in the broader context of India. We first examine the extent to which students in Hindi-medium schools benefited from the policy. Fig. 3 shows the year-wise average enrollment in grades 11 and 12 for schools under the Central Board of Secondary Education (CBSE) in West Bengal and Hindi medium schools under WBCHSE. The sample consists of an unbalanced panel of 233 CBSE schools in West Bengal and 159 Hindi medium schools under WBCHSE. CBSE is a national-level education board in India, overseeing both public and private schools, and is governed by the Government of India. During our period of study, CBSE has been conducting grade 12 examinations,[10] with question papers available in both Hindi and English. The raw data shows an increase in enrollment in Hindi medium schools after 2017 (post-policy change), while CBSE schools in West Bengal experienced a decline in enrollment, particularly in 2021. This trend is more pronounced in grade 11 compared to grade 12. This suggests that some students who were studying in Hindi medium schools under West Bengal Board of Secondary Education (WBSSE) up to grade 10 may have shifted to CBSE schools,[11] where exams are conducted in Hindi, to complete their higher secondary schooling (grades 11 and 12) prior to the policy change. It is plausible that the more motivated students would have migrated to CBSE schools. Since they are better students who decide to stay back in WBCHSE Hindi medium schools after the introduction of question papers in Hindi, we observe a statistically significant decrease

[10] CBSE conducts examinations for both grades 10 and 12.

[11] Students may also shift to neighboring states like Jharkhand, where the state board conducts examinations in Hindi or to CBSE schools in other states. We restrict ourselves to CBSE schools in West Bengal for availability of data.

in the share of repeaters, an increase in enrollment, and a decrease in the number of repeaters particularly for grade 11.

Our paper also has large potential policy implications in the context of India. Several state boards provide bilingual question papers for higher secondary education in India. Bilingual question papers are typically provided in English and the dominant regional language of each state, resulting in a disparity between the medium of instruction and the language used for test assessment. Similarly, several national (state) level exams[12] in India continue to hold their admission examinations to undergraduate education in English and Hindi (state vernacular). Students with limited proficiency in English or Hindi (state vernacular in case of state-level examinations) may be at a disadvantage in equitably engaging with and succeeding in such entrance examinations. The policy implication of our study could also have a bearing on labor market outcomes. The Union Public Service Commission (UPSC) conducts the Civil Services Exam,[13] where the question papers are provided in Hindi or English, although examinees can choose to write answers in English or one of the twenty-two recognized languages in the Indian constitution (including Hindi). While there has been discontent among the public regarding the non-availability of question papers in regional languages, whether it impacts labor market outcomes is worth exploring in future research for more inclusive language-based policies. Similar concerns also apply to members of the linguistic minority communities in states when the respective state government recruitment exams are conducted in dominant languages.

7. Conclusion

We find that language mismatch in terms of comprehending the question in a language different than the medium of instruction had significant impacts on school education even when there was no mismatch in the language in which the examinees could write their answers. Though the policy change might appear to be a minor one, the impacts on the targeted group are significant and meaningful. Moreover, the impact of the policy might be larger than we uncover in our results. While the only performance indicator that we consider is the share of repeaters, grade improvement as a result of the policy is not unlikely. However, we are constrained by data to investigate the impact of the policy on the second outcome. Moreover, the unavailability of student-level data limits our ability to identify the exact mechanism driving our results.

We believe that such language barriers are further compounded within the context of university education in India, where students often lack the option to choose their medium of instruction and are instead required to operate in a language in which they have limited proficiency. Future research should further explore the role of language in both school and higher education settings to inform policies aimed at making education more inclusive.

Declaration of competing interest

We have nothing to declare.

[12] Joint Entrance Examination (JEE)-Advanced is a national-level examination conducted only in Hindi and English, used for admission to premier engineering institutions in India, including the Indian Institutes of Technology (IITs). The IISER Aptitude Test (IAT) is also conducted in Hindi and English and serves as the entrance exam for undergraduate programs in natural and applied sciences at the Indian Institutes of Science Education and Research (IISERs) and the Indian Institute of Science (IISc). Common Law Admission Test (CLAT-UG) is another national-level test conducted only in English for admission into undergraduate law programs in national law universities.

[13] The Civil Services Examination, conducted by the UPSC, is used to recruit candidates for various prestigious administrative positions in the Government of India, including the Indian Administrative Service (IAS), Indian Police Service (IPS), and Indian Foreign Service (IFS).

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Appendix

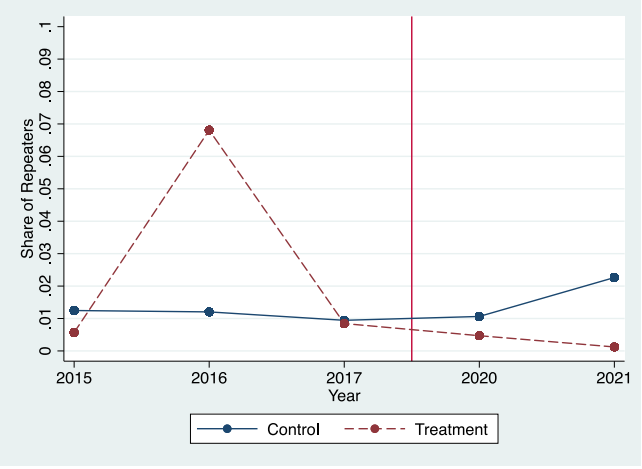


Fig. A.1. Year-wise average of share of repeaters by treatment status.
Notes: The above figure presents the share of repeaters over time for the control schools (Bengali/English medium) and the treated schools (Hindi medium). The vertical line denotes the implementation of the policy.

Table A.1
District-wise distribution of schools in the treatment and control group.

Districts	Control schools (1)	Treatment schools (2)
Alipurduar	27	3
Barddhaman	229	16
Haora	163	10
Hugli	52	3
Jalpaiguri	39	3
Kolkata	115	9
North 24 Pargana	275	9
Paschim Medinipur	306	2
Puruliya	106	1
Siliguri	30	6
Uttar Dinajpur	64	1
Total	1406	63

Notes: Column (1) reports the district-wise number of Bengali/English medium schools under WBCHSE in the control group and Column (2) reports the district-wise number of Hindi medium schools under WBCHSE in the treatment group.

Table A.2
Impact of question papers in Hindi on repeaters and enrollment (in levels).

	Share of repeaters (1)	No. of repeaters (2)	No. of enrollment (3)
Treat × Post	−0.03***	−13.84**	24.99**
	(0.011)	(5.656)	(10.259)
Mean Dep. Var (Control)	0.013	3.78	250
Observations	7345	7345	7345
Adjusted R^2	0.01	0.03	0.89
School controls	Yes	Yes	Yes
School FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Notes: Columns (1), (2) and (3) report the difference-in-difference estimates of the impact of the introduction of question papers in Hindi on the share of repeaters, number of repeaters and enrollment for grades 11 and 12 combined. The dependent variable in column (1) is defined as the number of repeaters as a ratio of the total enrollment. The sample consists of a balanced panel of 1406 Bengali/English medium schools under WBCHSE in the control group and 63 Hindi medium schools under WBCHSE in the treatment group. Treat is an indicator variable that takes a value of 1 for Hindi medium schools and 0 for Bengali/English medium schools. Post is an indicator variable that takes a value of 1 for the post-intervention years of 2020 and 2021 and 0 for the pre-intervention years of 2015, 2016, and 2017. All regressions include school-level controls that vary with time - electricity connection at the school, school accessibility by road, number of good classrooms in the school, number of functional toilets in the school, total number of teachers, and percentage of school teachers who are graduates. Column (2) additionally controls the total enrollment. All regressions include school and year fixed effects. Standard errors are clustered at the school level and reported in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

Table A.3
Placebo: impact of question papers in Hindi on share of repeaters.

	ln(Share of repeaters)		
	False treatment Year-2016 (1)	False treatment Year-2017 (2)	Grade 10 (3)
Treat × Post	0.026**	−0.017	−0.003
	(0.011)	(0.011)	(0.005)
Observations	4407	4407	7985
Adjusted R^2	0.02	0.01	0.04
School controls	Yes	Yes	Yes
School FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Notes: The dependent variable in all the regressions is the natural log of the number of repeaters as a ratio of the total enrollment. The sample in columns (1) and (2) includes only the pre-intervention years of 2015, 2016 and 2017 for grades 11 and 12 combined and consists of 1406 Bengali/English medium schools under WBCHSE in the control group and 63 Hindi medium schools under WBCHSE in the treatment group. Column (1) assumes that the fake treatment year is 2016 and therefore, the indicator variable Post takes a value of 1 for the years 2016 and 2017 and 0 for the year 2015. Column (2) assumes that the fake treatment year is 2017 and therefore, the indicator variable Post takes a value of 1 for the year 2017 and 0 for the years 2015 and 2016. The sample in column (3) includes all grade 10 schools, including schools that offer education up to grade 10. The sample includes a balanced panel of 1528 Bengali/English medium schools in the control group and 69 Hindi medium schools in the treatment group for each year. In column (3) Post is an indicator variable that takes a value of 1 for the post-intervention years of 2020 and 2021 and 0 for the pre-intervention years of 2015, 2016, and 2017. In all regressions, Treat is an indicator variable that takes a value of 1 for Hindi medium schools and 0 for Bengali/English medium schools. All regressions include school-level controls that vary with time - electricity connection at the school, school accessibility by road, number of good classrooms in the school, number of functional toilets in the school, total number of teachers, and percentage of school teachers who are graduates. All regressions include school and year fixed effects. Standard errors are clustered at the school level and reported in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

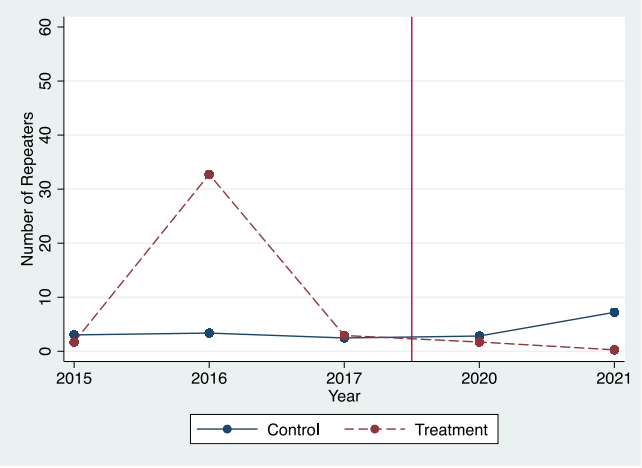


Fig. A.2. Year-wise average of number of repeaters by treatment status.
Notes: The above figure presents the average number of repeaters over time for the control schools (Bengali/English medium) and the treated schools (Hindi medium). The vertical line denotes the implementation of the policy.

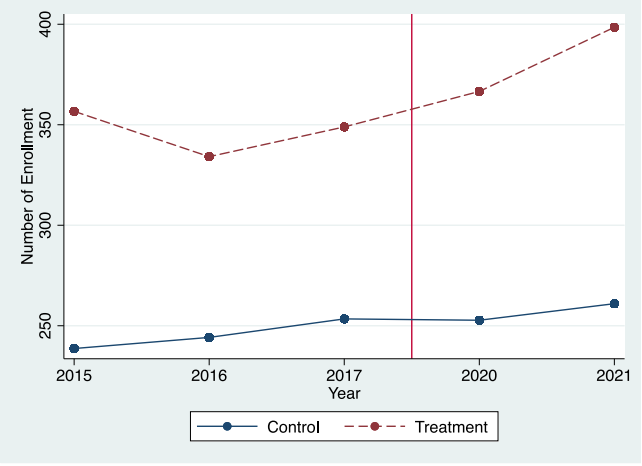


Fig. A.3. Year-wise average number of enrollment by treatment status.
Notes: The above figure presents the average number of enrollment over time for the control schools (Bengali/English medium) and the treated schools (Hindi medium). The vertical line denotes the implementation of the policy.

Data availability

Data will be made available on request.

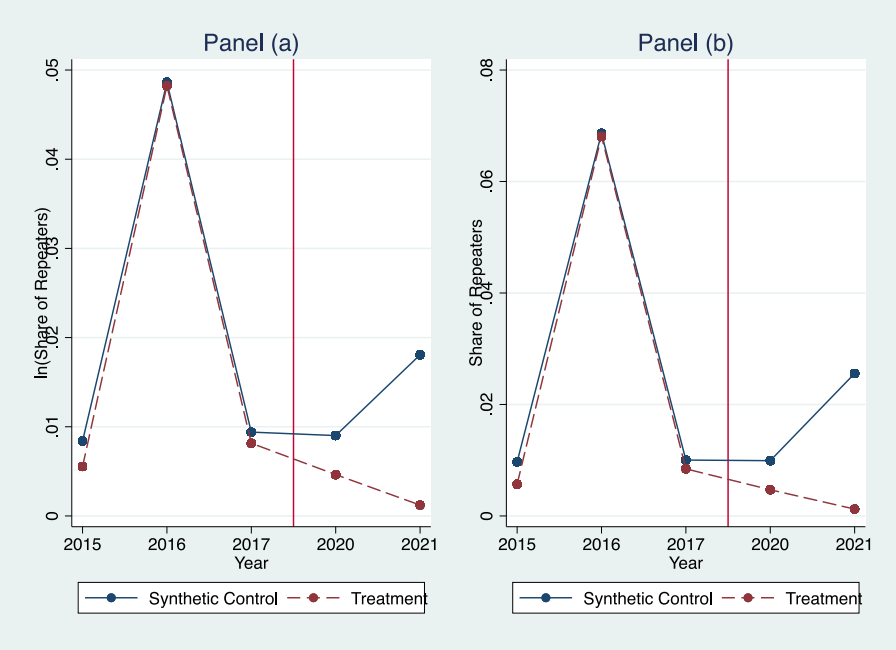


Fig. A.4. Synthetic difference-in-difference outcome plots.
Notes: The above figures present the time series of share of repeaters by treated schools (Hindi medium) and the synthetic control group (control schools selected via the procedure in Arkhangelsky et al. (2021)). Panel (a) presents annual averages of the share of repeaters and Panel (b) presents the annual average of the natural log of share of repeaters. Note that synthetic control group composition changes across panels to best match pre-treatment trends.

Table A.4
Comparison of school characteristics (mean) across medium of instructions.

Variables	Medium of instruction			
	Only English (N = 5)	Only Bengali (N = 1396)	Bengali and English (N = 5)	Only Hindi (N = 63)
	(1)	(2)	(3)	(4)
School characteristics				
Prop. of schools with electricity	1.00	0.99	1.00	0.98
	(0.00)	(0.09)	(0.00)	(0.14)
Prop. of schools approachable by all weather road	1.00	0.94	1.00	0.98
	(0.00)	(0.231)	(0.00)	(0.15)
Number of classrooms in good condition	16.96	12.76	23.60	12.04
	(16.55)	(7.35)	(12.86)	(8.93)
Number of functional toilets	19.56	6.19	11.60	8.60
	(28.29)	(4.21)	(3.98)	(8.21)
Number of teachers	15.20	25.51	39.96	20.84
	(6.79)	(9.04)	(12.97)	(9.82)
Percentage of teachers who are graduate	87.41	90.75	84.71	89.02
	(29.35)	(22.24)	(30.79)	(25.60)

Notes: Column (1) reports the summary statistics for only English medium schools. Column (2) reports the summary statistics for only Bengali medium schools. Column (3) reports the summary statistics for schools which has both English and Bengali as the medium of instruction. Column (4) reports the summary statistics of only Hindi medium schools. The sample consists of a balanced panel of 5 English medium schools, 1396 Bengali medium schools, 5 schools with both Bengali and English as the medium of instruction and finally 63 Hindi medium schools under WBCHSE across five years - 2015, 2016, 2017, 2020 and 2021. Standard deviations are reported in parentheses.

Table A.5
Robustness: impact of question papers in Hindi on share of repeaters.

	ln(Share of repeaters)	
	Only Bengali medium Schools as Control	Dropping Year 2017 from sample
	(1)	(2)
Treat × Post	−0.02***	−0.026**
	(0.007)	(0.011)
Observations	7320	5876
Adjusted R^2	0.01	0.01
School Controls	Yes	Yes
School FE	Yes	Yes
Year FE	Yes	Yes

Notes: The dependent variable in all the regressions is the natural log of the number of repeaters as a ratio of the total enrollment for grades 11 and 12 combined. The sample in column (1) includes a balanced panel of 1401 Bengali medium schools only in the control group and 63 Hindi medium schools under WBCHSE in the treatment group. Post is an indicator variable that takes a value of 1 for the post-intervention years of 2020 and 2021 and 0 for the pre-intervention years of 2015, 2016, and 2017. Treat is an indicator variable that takes a value of 1 for Hindi medium schools and 0 for Bengali medium schools. The sample in column (2) excludes the year 2017 and hence Post in this analysis takes a value 1 for the post-intervention years of 2020 and 2021 and 0 for the pre-intervention years of 2015 and 2016. This sample includes a balanced panel of 1406 English/Bengali medium schools in the control group and 63 Hindi medium schools in the treatment group. Treat is an indicator variable that takes a value of 1 for Hindi medium schools and 0 for Bengali/English medium schools. All regressions include school-level controls that vary with time - electricity connection at the school, school accessibility by road, number of good classrooms in the school, number of functional toilets in the school, total number of teachers, and percentage of school teachers who are graduates. All regressions include school and year fixed effects. Standard errors are clustered at the school level and reported in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

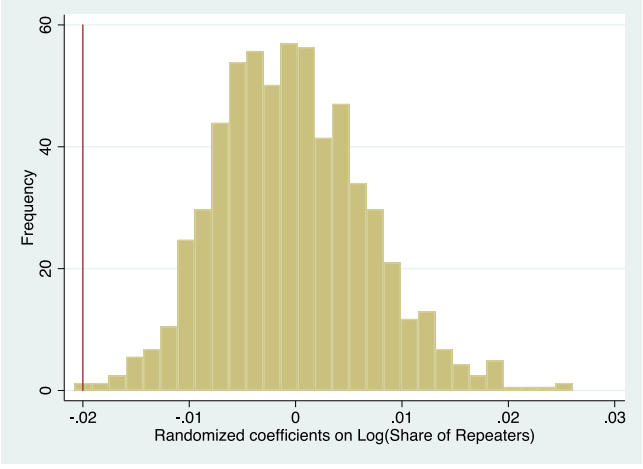


Fig. A.5. Exact randomization test.

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